R7.2: is any toy binary elliptic curve weak under GHS, end to end?

2026-11-01. Branch `research/r7-2-ghs-end-to-end`. Source: `RESULTS.md` (same directory); raw records in `raw/`; commands in `commands.sh`. Labels: measured = printed by `examples/r7_2_ghs_end_to_end.rs`; derived = arithmetic on measured values; extrapolated = a fitted line read outside the measured range.

Question. At a toy instance with N <= about 18, does Gaudry-style index calculus (IC) on the real GHS-descended Jacobian solve a planted ECDLP on $E(F_{q^n})$ with fewer counted group operations than Pollard rho, and than the best generic method, with relation collection, linear algebra mod ell and extraction all counted?

Result. No. 30 applicable instances, N = 12 to 28, 600 planted logs, every answer verified on E by all three methods. In bitop/v1 gate equivalents, expressed in E additions of the same curve (gae), the IC total divided by the best generic method (baby-step giant-step on 28 instances, tuned table rho on 2) lies between 1.91 [1.72, 2.11] (N = 28, q = 128, g = 4, n = 4) and 131 [123, 141] (95% bootstrap over the 20 paired logs); at N <= 18 the smallest is 9.18 [8.49, 9.96]. Against rho alone the smallest is 1.60 [1.25, 2.12]. For genus 4 over F_{q^4} the ratio halves with every doubling of q (11.5, 6.4, 3.3, 1.9 at q = 16, 32, 64, 128); that fit crosses 1 at N about 32 (31 to 33), **extrapolated**. For genus 4 over F_{q^3} the ratio stays at 18 to 30 (q = 16 to 256) and the linear algebra (q^2.1) outgrows the generic cost (q^1.4): no crossover is supported. Counting only Jacobian additions (one per E addition) would put IC below the generic method from N = 20 for n = 4, but that is a stage diagnostic: the library's Cantor addition costs 5.4 to 10.4 E additions.

The (l,n) = (3,4), g = 8, q = 8 run. Not hung and not a bug: R7.1's own example finishes it in 1175 s (point counts up to N_9 over $F_{2^{27}}$, about 9 min, and a 400000-element closure of 8 x 10^6 validated Cantor additions, about 14 min); IC on it costs 122 [104, 143] times BSGS.

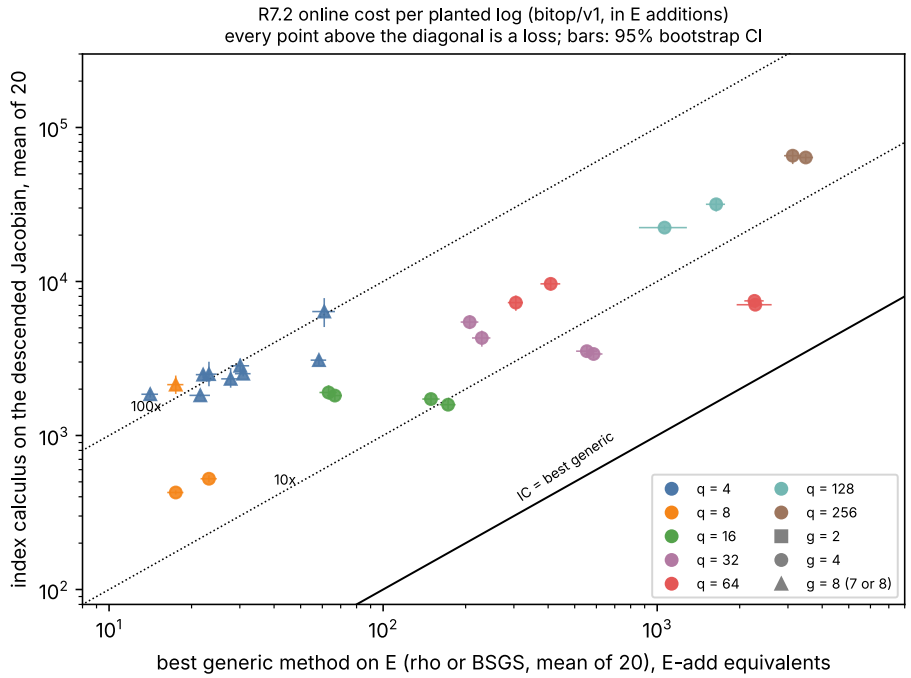


Figure 1: Online cost per planted log, IC against the best generic method on the same curve (E-add equivalents, mean of 20 paired logs, 95% bootstrap bars). Source: `raw/run_*.json`, `figure_data.tsv`.

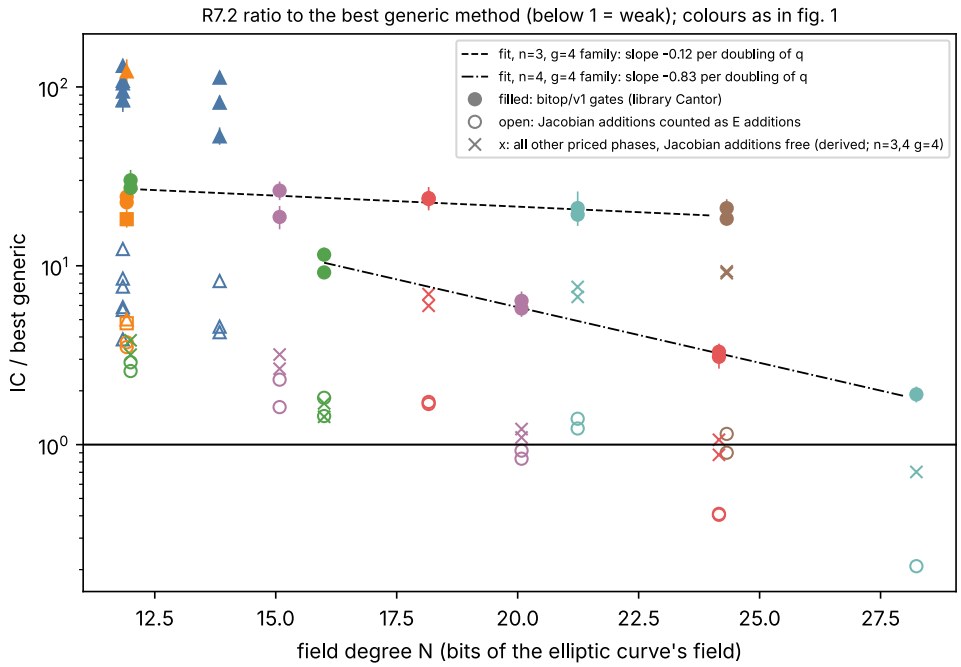


Figure 2: Ratio IC / best generic against N. Filled: bitop/v1 gates with the library's Cantor arithmetic; open: Jacobian additions counted as E additions (stage diagnostic); x: all other priced phases with Jacobian additions free (derived). Dashed: log-log fits of the two genus-4 families.

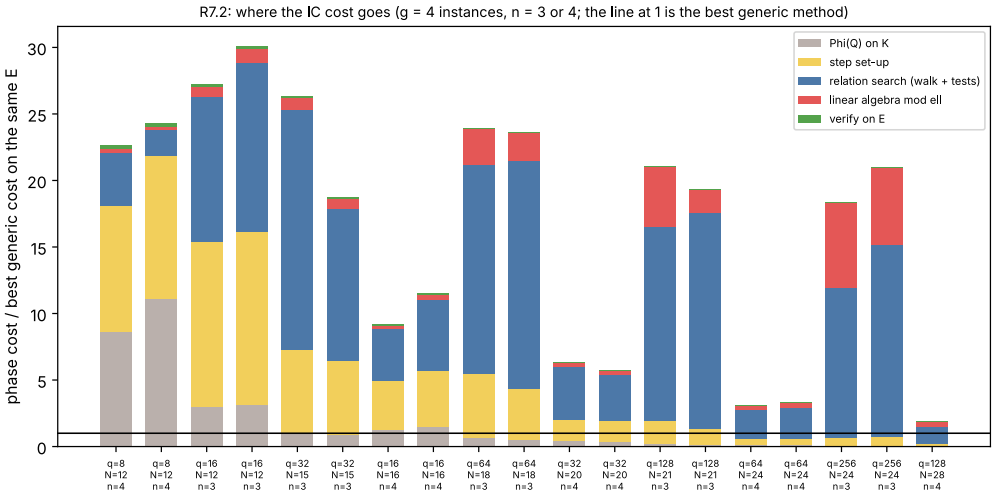


Figure 3: Where the IC cost goes (mean of 20 logs), as a multiple of the best generic cost on the same curve.

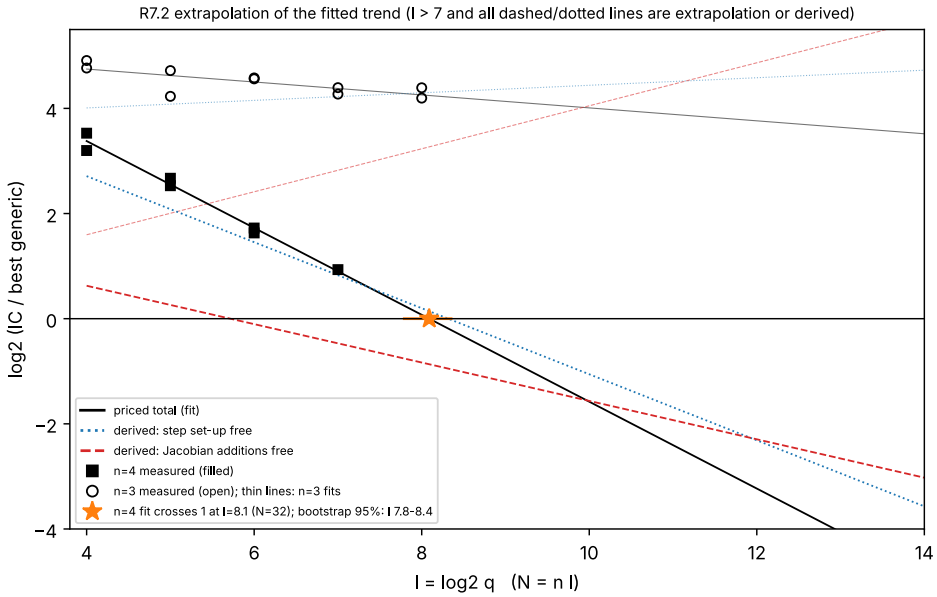


Figure 4: Fitted trend of the two genus-4 families extended past the measured sizes (extrapolation; dashed and dotted lines are derived variants). Fit and bootstrap: `raw/fit_genus4.json`.

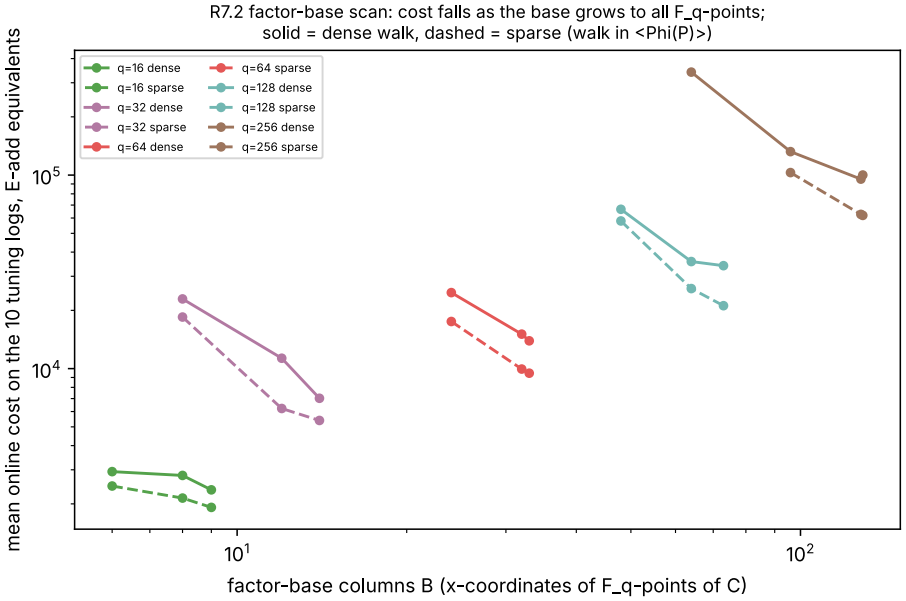


Figure 5: Factor-base scan on the 10 tuning logs (first curve of each q, n = 3, g = 4): the cost falls as the base grows to all F_q-points; sparse relations beat dense ones.

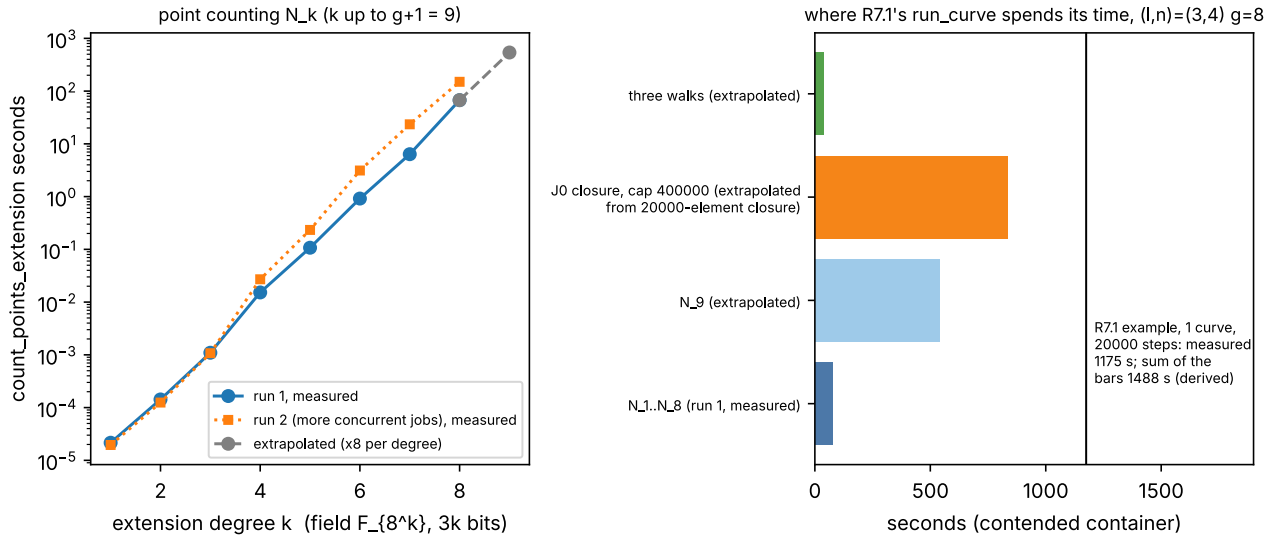


Figure 6: The (l,n) = (3,4), g = 8, q = 8 run of R5.2/R7.1: point-counting time by extension degree (two profile runs) and the phases of R7.1's run_curve against its measured total. Source: `raw/profile_*.json`, `raw/r71_rerun.log`.

All instances

Online cost in E-add equivalents, mean of 20 holdout logs; ratio = IC / best generic with 95% bootstrap interval; J/E = Jacobian additions over the best generic method's E operations (stage diagnostic).

instance (l,n,a,beta)	N	q	g	ell	IC	generic	ratio (95% CI)	J/E
ghs-l2-n6-a0-b0x228	12	4	8	29	1852	14.1	131.10 (122.81, 141.08)	5.66
ghs-l2-n6-a0-b0x221	12	4	8	67	1822	21.5	84.66 (78.07, 92.34)	3.88
ghs-l2-n6-a0-b0xe2	12	4	8	83	2502	23.1	108.11 (90.58, 129.32)	8.48
ghs-l2-n6-a0-b0x28	12	4	8	131	2332	27.8	83.95 (72.53, 99.16)	5.87
ghs-l2-n6-a0-b0x220	12	4	8	149	2835	30.1	94.20 (83.96, 105.23)	7.65
ghs-l2-n6-a0-b0x229	12	4	8	1049	6379	61.0	104.62 (82.13, 129.95)	12.41
ghs-l3-n4-a0-b0xa2	12	8	2	7	117	6.4	18.22 (16.35, 20.58)	4.77
ghs-l3-n4-a0-b0x3	12	8	8	37	2139	17.5	122.14 (104.47, 142.82)	5.01
ghs-l3-n4-a0-b0x8	12	8	4	37	426	17.5	24.35 (22.79, 26.12)	3.51
ghs-l3-n4-a0-b0x16	12	8	4	83	524	23.1	22.63 (20.65, 24.73)	3.73
ghs-l4-n3-a0-b0x36f	12	16	4	1013	1816	66.7	27.24 (24.31, 30.56)	2.58
ghs-l4-n3-a0-b0x74b	12	16	4	1039	1902	63.2	30.07 (26.65, 34.36)	2.89
ghs-l2-n7-a0-b0x4d	14	4	7	79	2483	22.0	112.67 (103.69, 122.53)	8.22
ghs-l2-n7-a0-b0x4c	14	4	8	157	2519	30.9	81.62 (75.11, 89.46)	4.25
ghs-l2-n7-a0-b0x58	14	4	8	677	3085	58.3	52.91 (47.54, 59.36)	4.57
ghs-l5-n3-a1-b0x31a6	15	32	4	16349	4298	229.2	18.76 (16.00, 21.59)	1.62
ghs-l5-n3-a1-b0x1df9	15	32	4	16451	5451	207.0	26.34 (23.29, 29.62)	2.31
ghs-l4-n4-a1-b0x3d2c	16	16	4	8179	1725	149.4	11.54 (10.57, 12.63)	1.83
ghs-l4-n4-a1-b0x2944	16	16	4	8191	1585	172.6	9.18 (8.49, 9.96)	1.44
ghs-l6-n3-a0-b0x37711	18	64	4	32801	7291	305.0	23.91 (20.42, 27.56)	1.69
ghs-l6-n3-a1-b0x1e702	18	64	4	65563	9652	408.7	23.61 (21.11, 26.40)	1.73
ghs-l5-n4-a0-b0x48eb3	20	32	4	131023	3528	553.9	6.37 (5.60, 7.18)	0.93
ghs-l5-n4-a0-b0x90f2d	20	32	4	131129	3382	585.9	5.77 (5.18, 6.50)	0.83
ghs-l7-n3-a0-b0x11ceab	21	128	4	524221	22375	1063.9	21.03 (17.54, 26.04)	1.40
ghs-l7-n3-a1-b0x1dd060	21	128	4	1047199	31714	1641.5	19.32 (16.70, 21.76)	1.23
ghs-l6-n4-a0-b0x5716d4	24	64	4	2097383	7064	2280.3	3.10 (2.66, 3.67)	0.41
ghs-l6-n4-a1-b0xb1261a	24	64	4	2097803	7484	2264.7	3.30 (3.02, 3.62)	0.41
ghs-l8-n3-a1-b0xc136f2	24	256	4	4194973	65568	3122.2	21.00 (18.37, 23.55)	1.15
ghs-l8-n3-a0-b0x2a521b	24	256	4	4195349	63833	3483.9	18.32 (16.84, 19.92)	0.90
ghs-l7-n4-a1-b0x19e70cc	28	128	4	33553901	17129	8969.6	1.91 (1.72, 2.11)	0.21

Cost model, protocol and limits

Unit: bitop/v1 (`docs/units/BITOPS.md`); Jacobian-side field operations are recorded by the library's operation ledger while the Cantor arithmetic runs (`add_unchecked`, no re-validation), E-side group operations by formula (ledger check: 1.000 over 400 additions); arithmetic mod ell is counted for the non-zero entries. Per instance: 10 tuning logs (relation mode, factor-base size, rho multiplier width) and 20 fresh paired holdout logs. Sparse relations chosen in 30 of 30 instances; the whole F_q-point set as factor base in 26 of 30. Limits: toy sizes; generic Cantor arithmetic; no large-prime variation; table-based generic arms without the negation map; ecbench not used (no replay certificate, no scoreboard entry); cover construction unpriced; a selected class of curves (beta of magic number 3 for n = 4); N = 32 not buildable with the library's irreducible-polynomial table (degree <= 31); wall times contended and secondary. No claim about any deployed curve.